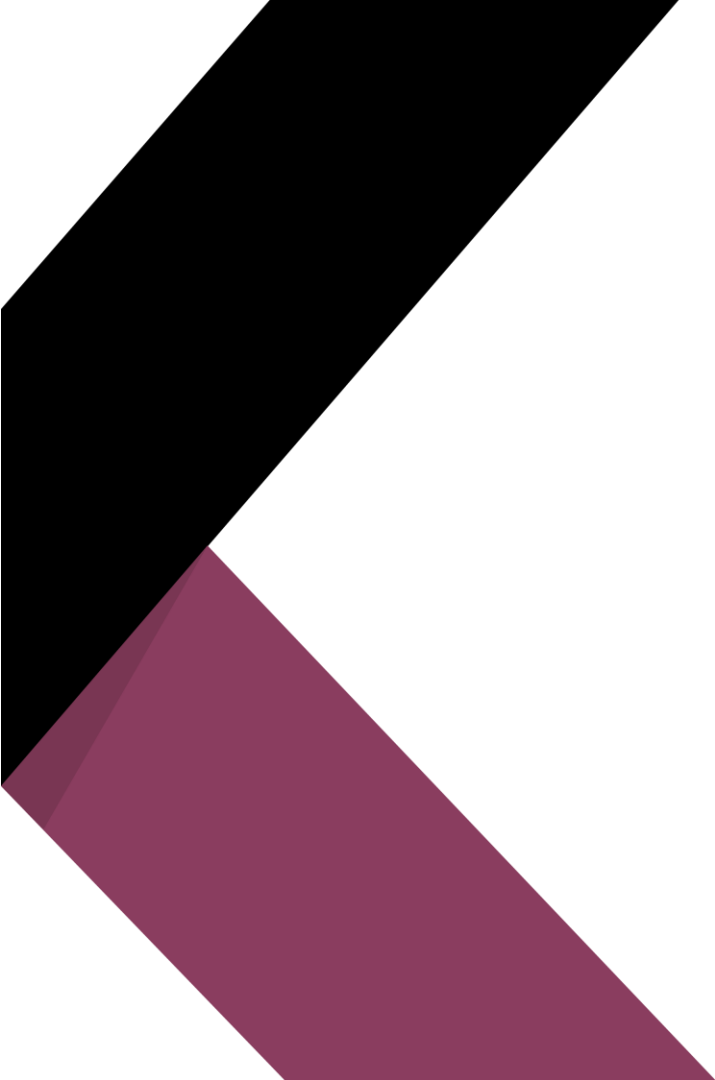


A large black triangle pointing downwards and a smaller maroon triangle pointing upwards, both located on the right side of the slide.

Sensors&Actuators



Sensors

Basic Definitions

Sensor

Transducer

- **Sensor:** the primary element of a measurement chain that converts the input variable into a signal suitable for measurement
- **Transducer:** A device that accepts information in the form of a physical or chemical input variable and converts it into an output variable of the same or different nature
- **Transmitter:** A device that receives a measurement variable and produces a normalized output signal



Sensor vs Transducer

The sensor is the first element in the measurement chain.

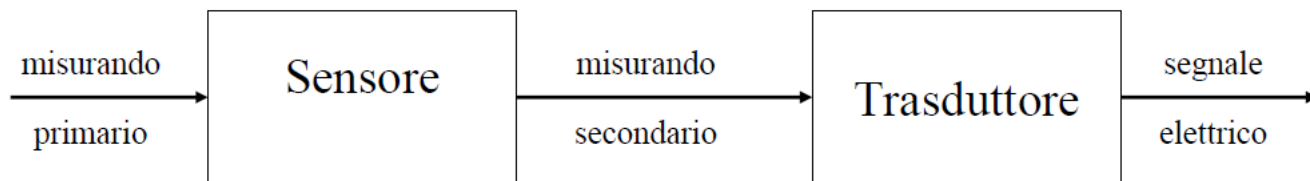
It has the task of converting the physical quantity to be measured (by measuring) into another more easily manageable.

The transducer is a sensitive device that provides a measurable electrical signal in response to a specific measurand.

A **transducer** is a sensor but a **sensor** is not necessarily a transducer



Sensor vs Transducer





Sensor Classification

- Active/Passive
- According to the measured quantity: sensors for temperature, humidity, illuminance, speed, pressure, ...
- Based on the quantity they provide at the output: resistive, inductive, capacitive,
- Voltage, Current, ...
- Analog / digital



Active vs Passive

- Active if the physical effect on which it is based ensures the transformation of the measurand's own energy into electrical energy (thermal, mechanical, irradiation, ...).
- Examples: Thermoelectric (thermocouple), Pyroelectric (crystals whose polarization depends on the temperature), ...

Passive if the effect of the measurand results in a change in the impedance of the sensitive element.

Examples: strain gauges, magnetic, ...



Conditioning Circuit

- **Passive S / T:** the conditioning circuit is essential for the generation of the electrical signal (assembly).
- **Active S / T:** the conditioning circuit has the task of adapting the parameters of the electricity generated by the transducer to the input characteristics of the measurement system (signal conditioning).



Main Features

S/T

- **Measuring**: quantity to be measured.
- **Measurement** principle: physical principle on which the generation of the output signal is based.
- **Significant properties**: type of sensitive element, type of construction, internal circuitry, ...
- **Range**: upper and lower limit of variation of the measurand.



- **Species:** nature of the output quantity.
- **Normal** operating range (output range): range of output values when the input varies in the input range.
- **Deliverable power:** limit value of the power that the sensor can supply to the downstream user system. If the output is current, the load impedance is specified.
- **Output impedance**
- **Uncertainty of output:** width of the band including all the values that could be assumed, with a certain probability (level of confidence), to represent the value of the output corresponding to a certain operating condition.



- **Conversion function:** function that allows you to obtain the output value from the input.
- **Calibration constant:** slope of the calibration curve, if it is linear;
- **Calibration uncertainty:** width of the range of values.



Statical Features

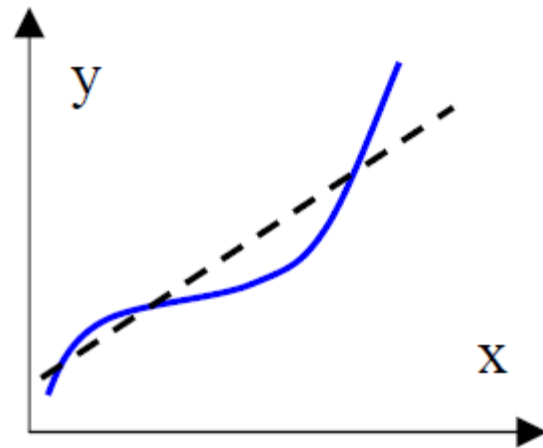
- **Sensitivity:** slope of the conversion curve at a certain point and corresponds to the inverse of the slope of the calibration curve.
- **Stability:** ability to maintain operating characteristics unaltered over time (long and short term)

$$S = \frac{dy}{dx}$$



Statical Features

- **Linearity:** indicates how much the calibration curve differs from the straight line. It is the maximum deviation from a line.





Statical Features

Resolution: variation in the value of the measurand which causes an appreciable variation in the value of the output quantity. If the sensor works close to zero, it is called a threshold.

Repeatability: aptitude of the instrument to provide slightly different values of the output quantity, when the same is applied to the input by measuring, under the same operating conditions. It is expressed in a similar way to the calibration uncertainty.

Hysteresis: the maximum difference between the values of the output corresponding to the same measurand, when all the values of the measuring range are considered, and each value is reached with first increasing and then decreasing measuring.



- Frequency domain:
 - **Bandwith.**
- Dominio del tempo:
 - **T rise and T fall of the output signal**
 - **Time Constant**
 - **Settling time**



Sensors

Example

Physical principle	Typical application	Measurand	Output
Resistive The variation of the sensing element electric resistance depends on the measurand.	Thermistor or resistance thermometer Potentiometer Hot-wire anemometer Resistive hygrometer Chemoresistor	Temperature Displacement, force, pressure Flow Humidity Presence of gas	Change in resistance
Capacitive The sensing element capacitance depends on the measurand.	Parallel-plate capacitor sensor Rotary-plate capacitor sensor Differential capacitor Capacitance manometer Humidity sensor Capacitive diaphragm	Displacement, force, liquid level, pressure Displacement, force, angular position, torque Small displacement Very low pressure Moisture Pressure	Capacitance or change in capacitance
Inductive The sensing element inductance depends on the measurand.	Linear variable differential transformer Self inductance sensor Eddy current sensor	Displacement, torque Displacement, torque, liquid level Position, conductivity, thickness, cracks in materials	Inductance or change in inductance
Reluctive The variation in the reluctance path between two or more coil depends on the measurand.	Linear variable differential transformer Rotary variable differential transformer Microsyn Resolver Synchro Reluctive diaphragm	Linear displacement Angular rotation Angular displacement Position Position, torque Pressure	Voltage Voltage Voltage Voltage Voltage Change in reluctance
Electromagnetic In any circuit capturing a magnetic flux, whenever the flux changes an electromotive force is induced. (Faraday law)	Linear velocity sensor Flowmeter Tachometer generator Torque sensor	Linear velocity Flow Angular speed Torque	Voltage
Piezoresistive effect Resistance of the sensing element depends on the strain.	Strain gauge	Stress, strain, Fluid pressure, displacement, force	Change in resistance
Hall effect If the sensing element, carrying current, is put in a magnetic field a differential in electric potential among its sides is generated.	Gaussmeter Wattmeter	Magnetic field, displacement Power	Voltage



Sensors

di sensori

Magnetoresistive effect Resistance of the sensing element depends on the strain.	Magnetoresistor	Magnetic field, linear and angular displacement, proximity, position	Change in resistance
Piezoelectric effect Subjecting the sensing element to stress there is a generation of electric charge.	Vibration cables Active and passive force sensor Piezoelectric microphone Piezoelectric temperature sensor	Vibration Force Ultrasonic waves Temperature	Voltage or charge
Pyroelectric effect The sensing element generates an electric charge in response to a heat flow.	Heat flowmeter Pyroelectric sensor	Change in the temperature	Voltage
Thermoelectric effect When there is a difference in temperature between two junctions of different metals, a difference of electric potential is generated.	Thermocouples, thermopiles, infrared pyrometer	Difference of temperature	Voltage
Ionization effect The sensing element when exposed to the measurand becomes ionized.	Electrolytic sensor Vacuum gages Chemical ionizer	Electrical conductivity, pH Pressure Atomic radiation	Current
Photoresistive The electric resistance of the sensing element is caused by the incidence of optical radiation.	Photoresistor, photodiode, phototransistor, photofet	Light, position, motion, sound flow, force	Change in resistance
Photovoltaic effect When the sensing element is subject to a radiation it generates an electric potential	Flame photometer Light detector Pyrometers	Light intensity Light, position, motion, sound flow, force Temperature	Voltage
Acoustooptic effect The interaction of an optical wave with an acoustic wave produces a new optical wave	Acoustic optic deflection, Bragg cell	Physical vibration	Phase modulated voltage signal
Doppler effect The apparent frequency of a wave train changes in dependence of the relative motion between the source of the train and the observer.	Remote sensor of linear velocity, Doppler radar, laser Doppler velocimeter	Relative velocity	Frequency
Thermal radiation An object emanates thermal radiation, which intensity is related to its temperature	Pyrometer	Temperature	Voltage



Actuators



Actuator Definition

Device that, in an automation system, transforms an automatic command decision, processed by an electronic control board, into a physical action on the process being regulated.

An actuator therefore allows you to act on a system in order to modify its behavior and obtain the desired output.

More generally, the actuator can be defined as a transducer capable of transforming a signal from a physical domain (typically of the electrical type) into an equivalent quantity belonging to another domain (e.g., mechanical, hydraulic, thermal, etc.)



- Classification according to the type of physical quantity into which the command signal is transduced. The most common ones are:
 - Mechanical
 - Plumbers
 - Electrical
- Classification according to the type of electric actuation command:
 - Digital;
 - Analog (PWM)



Digital Actuator Relay

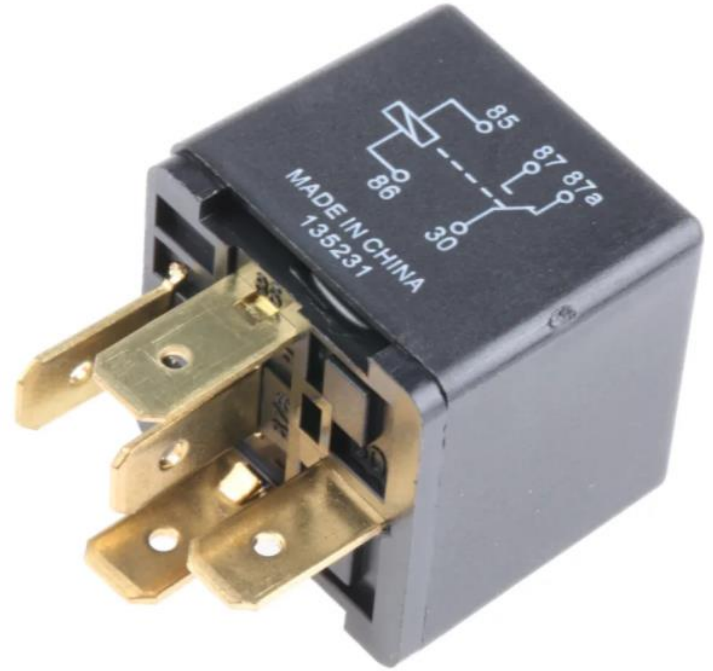
The relay is one of the simplest and most used actuators in the automotive field.

The relay can be defined as an electromechanical switch that allows you to divide the control of an element from the power part necessary for implementation.

The relay therefore consists of two distinct circuits:

Control circuit: consists of a coil; when this coil is crossed by electric current, it is excited

Power circuit: consists of the contacts and the user; the contacts can be normally open or normally closed





Relay **HSD**

- An **HSD** relay (High Side Driver Switch) is a relay which in order to be commanded "active", the control unit that manages its control, needs to use a "high potential".
- As a result, the control coil will have one of the two pins connected to a low reference potential

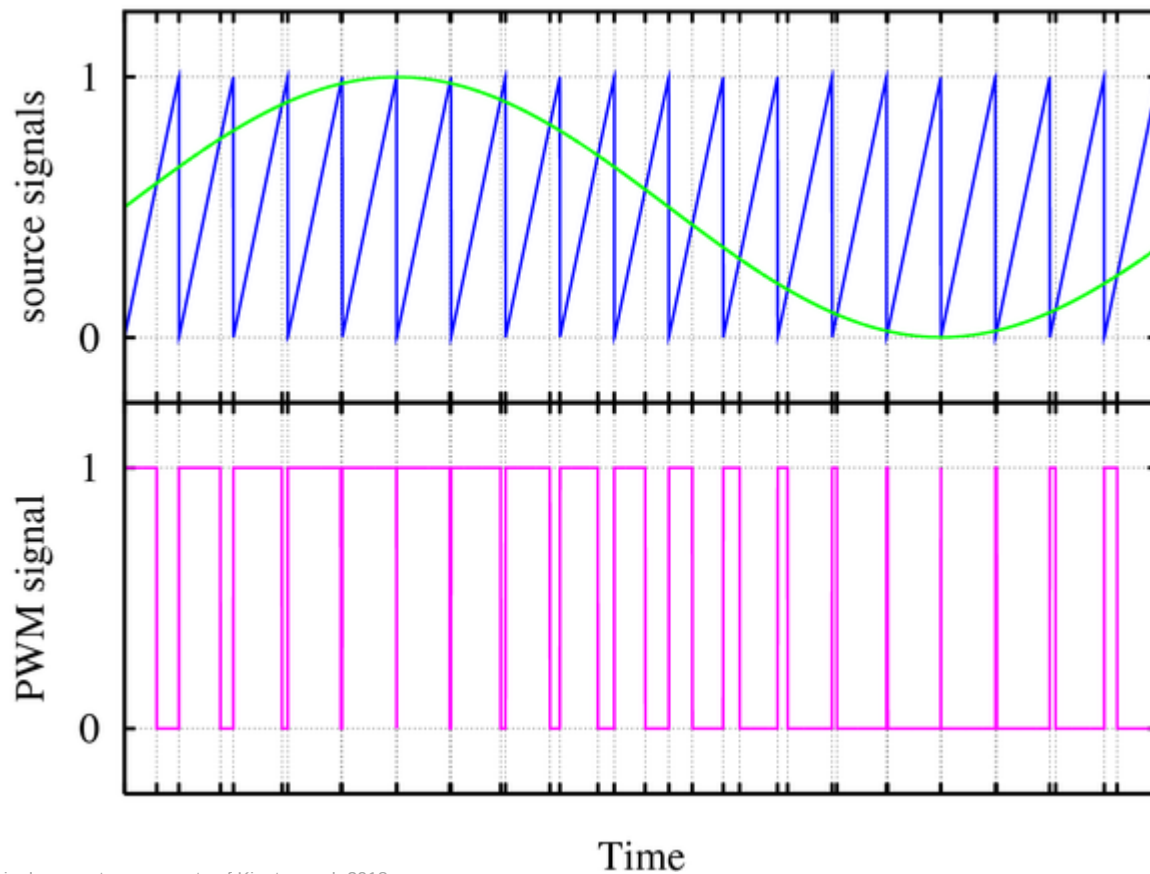


Relay **LSD**

- **LSD** relay (Low Side Driver Switch) is a relay which in order to be commanded "active", the control unit that manages its control, needs to use a "low potential"
- As a result, the control coil will have one of the two pins connected to a high reference potential

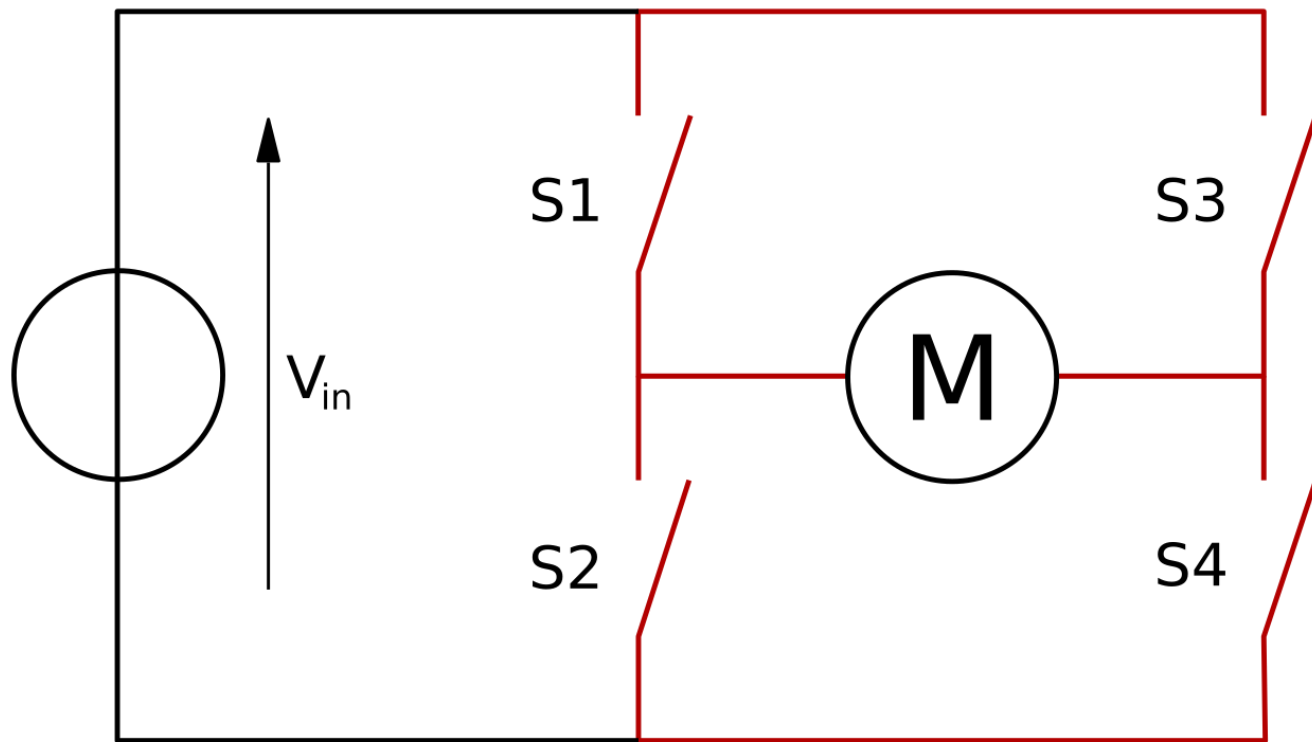


Command PWM





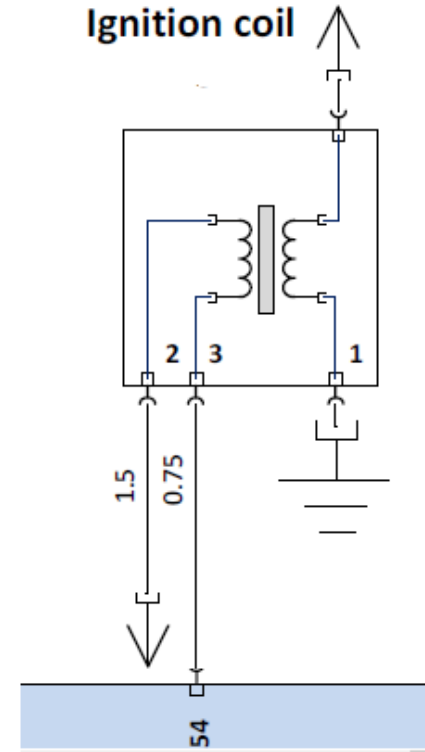
H - Bridge





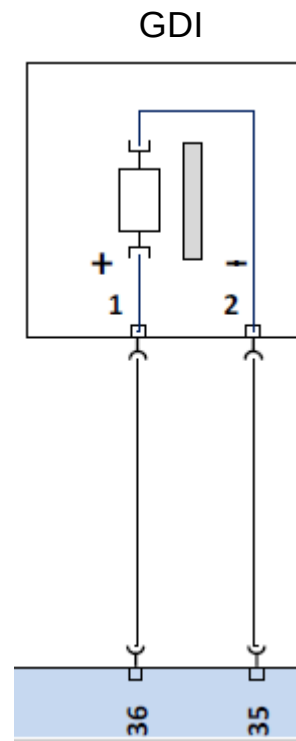
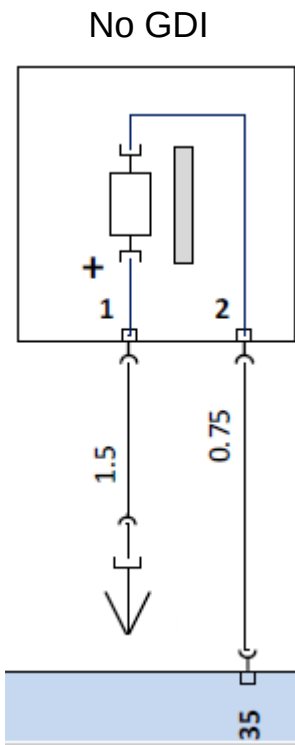
Use Case Ignition Coil

- Need to generate an "ignition spark" photovoltaic arc
- Therefore, it is necessary to generate a high voltage through the use of a transformer
- Need to use a primary power source that is alternating





Use Case Fuel Injector



CONTACTS

Bruno D'Angelo

[Bruno.dangelo@kineton.it](mailto:bruno.dangelo@kineton.it)

Antonio Vitale

antonio.vitale@kineton.it

Via E. Gianturco, 23
(presso complesso Mecfond)
80146 Napoli

www.kineton.it

